

Sheikh Abdur Raheem Ali

<http://abdur-raheem.com>

Email : ali@abdur-raheem.com

Mobile : +1 (604) 790 5369



PROFESSIONAL EXPERIENCE

- **Independent** Toronto, ON, Canada
Research Engineer May 2024 – Present
 - Sabbatical to travel, learn transformer interpretability, and study alignment of large language models.
- **Microsoft Corporation** Vancouver, BC, Canada
Software Engineer II Apr 2021 – May 2024
 - Co-owned impersonation protection in Defender for Office 365. Worked with the FBI, NYPD, U.S DoD to protect 250m+ people against phishing threats. Cut mailbox intelligence false positive rates by 55%.
 - Reduced download time for Substrate (the #2 largest monorepo in MS) from 2-3 hours to ≈ 20 minutes. Used by over 10% of all developers at the company and saves at least \$1m per annum.

Software Engineering Intern Jul. 2020 – Apr 2021
 - Built priority accounts. Was asked to stay on contract after the end of my internship until I graduated.
- **Dubai Electronic Security Center** Cyber Intelligence Lab, University of Dubai, Dubai, UAE
ML Research Intern Dec 2019 – Jan 2020
 - Recruited by the Government's Head of Research & Innovation. Did a 5-week project in my first weekend.

EDUCATION

- **Stanford University** Palo Alto, CA, USA
Graduate Coursework in Advanced Software Systems Dec 2020 – Apr 2021
- **American University of Sharjah** Sharjah, UAE
Bachelor of Science in Computer Science; Minor in Applied & Computational Math Sep 2017 – Dec 2020

SELECTED PUBLICATIONS

- **Patterns and Mechanisms of Contrastive Activation Engineering:**
Yixiong Hao, Ayush Panda, Stepan Shabalin, **Sheikh Abdur Raheem Ali**
International Conference on Learning Representations (ICLR) 2025
- **Interpreting Large Text-to-Image Diffusion Models with Dictionary Learning:**
Stepan Shabalin, Dmitrii Kharlapenko, Yixiong Hao, Ayush Panda, **Abdur Raheem Ali**, Arthur Conmy
Conference on Computer Vision and Pattern Recognition (CVPR) 2025
- **Scaling Laws for Activation Steering with Llama 2 Models and Refusal Mechanisms:**
Sheikh Abdur Raheem Ali, Justin Xu, Ivory Yang, Jasmine Xinze Li, Ayse Arslan, Clark Benham
International Conference on Machine Learning (ICML) 2025

AWARDS & SCORES

- **2nd place (USD 35,000) @ HITB⁺ CyberWeek AI Challenge:** Captain of a team including students from Harvard and UToronto that used genetic algorithms to modify existing malware to defeat virus detection agents.
- **Full Scholarship @ European Summer Program on Rationality:** Invited to the UK to attend seminars on topics related to applied mathematics, AI safety, and cognitive science. Taught by ex-Google employees and MIT graduates.

TEACHING AND MENTORING EXPERIENCE

- **Mentor @ Berkeley AI Safety Initiative:** Lead for Supervised Program in Alignment Research in Spring 2024. My students' project was accepted as a poster at Technical AI Safety Conference 2024.

JUDGING AND REVIEWING

- **Judge @ Georgia Tech AI Safety Initiative:** Jury member scoring projects in the AI ATL hackathon.
- **Reviewer @ NeurIPS 2025:** Reviewer for the conference.